

Name: \_\_\_\_\_

Teacher: \_\_\_\_\_

# Gosford High School

## 2013

### Year 11 Preliminary

### Mathematics Extension 1

### Assessment Task #1

#### INSTRUCTIONS:

TIME: 60 minutes + 5 minutes reading time

- Write using blue or black pen.
- Board approved calculators may be used.
- In questions 5 – 7, show relevant mathematical reasoning and/or calculations

|                 |                     |            |
|-----------------|---------------------|------------|
| Questions 1 – 4 | Multiple Choice     | /4         |
| Question 5      | Algebra             | /14        |
| Question 6      | Inequalities        | /8         |
| Question 7      | Counting Techniques | /14        |
| <b>TOTAL</b>    |                     | <b>/40</b> |

Question 1:

Russel is answering an inequality and writes the answer

$$-2 \leq x < 5 \text{ and } x \geq 1$$

This statement simplifies to:

a)  $1 \leq x \leq 5$

b)  $1 < x \leq 5$

c)  $1 \leq x < 5$

d)  $1 < x < 5$

Question 2:

Evaluate  $2 \times \frac{3^{n+1}}{3^{n-1}}$

a) 1

b) 18

c) 2

d) 36

Question 3:

A committee of  $n$  people is to be chosen from  $n + 1$ . Which of the following is **NOT EQUAL** to the number of committees possible:

a)  ${}^{n+1}C_n$

b)  ${}^{n+1}P_n \div n!$

c)  ${}^{n+1}P_n$

d)  $n + 1$

Question 4:

A bag contains 1 white and 5 black marbles. 5 will be taken from the bag and placed in a line. How many distinct arrangements are possible.

a) 6

b) 5

c)  ${}^6P_5$

d)  $5!$

Question 5:

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- a) Fully factorise:
- i)  $(x + y)^2 - 2x - 2y$  (2)
  - ii)  $a^6 - b^6$  (2)
  - iii)  $(x^2 + xy)^2 - (xy + y^2)^2$  (2)
- b) Simplify  $\frac{4^m \times 27^{m-n}}{6^{2m}}$  (2)
- c) By factorising, find the result of dividing  $2^{n+3} + 2^n$  by 3 (2)
- d) If  $x + \frac{1}{x} = 3$  (Do not attempt to find  $x$ )
- i) Show that  $x^2 + \frac{1}{x^2} = 7$  (2)
  - ii) Hence find the value of  $x^3 + \frac{1}{x^3}$  (2)

Question 6:

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- a) Solve  $\frac{2}{x+2} < 5$  (2)
- b) Solve  $\frac{x+2}{(x-5)(x+4)} \geq 0$  (2)
- c) Find all positive values of  $x$  for which  $\frac{6}{x} > x - 1$  (2)
- d) Solve  $\frac{x+3}{x^2+6x+9} < 0$  (2)

Question 7:

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- a) i) How many 7 letter words can be made from the letters of **EVEREST**? (1)  
ii) How many have the **E**'s together? (1)
- b) 4 distinct Shrek and 3 different Toy Story movies are put on a shelf. How many arrangements are possible if:  
i) there are no restrictions (1)  
ii) no two movies from the same series can be adjacent (1)
- c) 6 Women and 3 men sit at a circular table with 9 seats. In how many ways can this happen if:  
i) there are no restrictions (1)  
ii) the men all sit together (1)  
iii) 2 specific women must not sit together (2)
- d) A committee of 6 students and 3 teachers is to be selected from 25 students and 10 teachers. In how many ways can this happen if:  
i) there are no restrictions (1)  
ii) a certain teacher, the principal must be on the committee (1)  
iii) a certain student and her boyfriend cannot both be on the committee (2)
- e) Solve for  $n$ :  
$${}^nC_5 \div {}^nC_6 = \frac{2}{3}$$
 (2)

# 2013 Prelim Ext 1

## Task 1

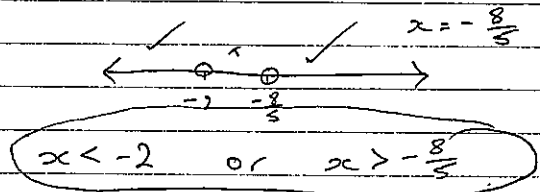
C  
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C  
A

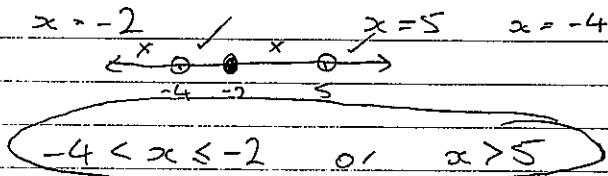
a) i)  $(x+y)^2 - 2(x+y)$   
 $= (x+y)(x+y-2)$   
 ii)  $(a^3-b^3)(a^3+b^3)$   
 $= (a-b)(a^2+b^2+ab)(a+b)(a^2-ab+b^2)$   
 iii)  $(x^2+xy+xy+y^2)(x^2+xy-xy-y^2)$   
 $= (x^2+2xy+y^2)(x^2-y^2)$   
 $= (x+y)^2(x-y)(x+y)$   
 $= (x+y)^3(x-y)$

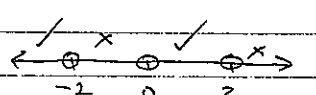
b)  $\frac{2^{2n} \times 3^{3n-3n}}{2^{2n} \times 3^{2n}} = 3^{n-3n}$

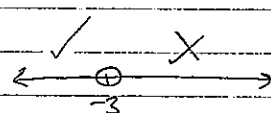
c)  $\frac{2^{n+3} + 2^n}{3} = \frac{2^n(2^3 + 1)}{3}$   
 $= 3 \times 2^n$

d) i)  $(x + \frac{1}{x})^2 = 3^2$   
 $x^2 + 2 + \frac{1}{x^2} = 9$   
 $x^2 + \frac{1}{x^2} = 7$   
 ii)  $x^3 + \frac{1}{x^3} = (x + \frac{1}{x})(x^2 - 1 + \frac{1}{x^2})$   
 $= 3 \times (7-1)$   
 $= 18$

b) a) CP:  $x+2=0$   $2=5x+10$   
 $x=-2$   $-8=5x$   
 $x=-\frac{8}{5}$   
  
 $x < -2$  or  $x > -\frac{8}{5}$

b) CP:  $x=-2$   $x=5$   $x=-4$   
  
 $-4 < x \leq -2$  or  $x > 5$

c) CP:  $x=0$   $6=x^2-x$   
 $x^2-x-6=0$   
 $(x-3)(x+2)=0$   
 $x=3$  or  $-2$   
  
 $x < -2$  or  $0 < x < 3$   
 not positive.

d) CP:  $x=3$   $x^2+6x+9=0$   
 $(x+3)^2=0$   
 $x=-3$   
  
 $x < -3$

$$7) a) i) \frac{7!}{3!}$$

$$ii) 5!$$

$$b) i) \frac{7!}{4! \times 3!}$$

$$ii) 4! \times 3!$$

$$c) i) 8!$$

$$ii) 6! \times 3!$$

$$iii) 8! - 7! \times 2!$$

$$d) i) {}^{25}C_6 \times {}^{10}C_3$$

$$ii) {}^{25}C_6 \times {}^9C_2$$

$$iii) {}^{25}C_6 \times {}^{10}C_3 - {}^{23}C_4 \times {}^{10}C_2$$

$$e) \frac{{}^nC_5}{{}^nC_6} = \frac{2}{3}$$

$$\frac{n!}{(n-5)!5!} \div \frac{n!}{(n-6)!6!} = \frac{2}{3}$$

$$\frac{n!}{(n-5)(n-6)!5!} \times \frac{(n-6)!6 \times 5!}{n!} = \frac{2}{3}$$

$$\frac{6}{n-5} = \frac{2}{3}$$

$$18 = 2n - 10$$

$$n = 14$$